
PETalytics: Using Staged Semi-Supervised Learning On ESM-2 Embeddings to Fight Plastic Pollution

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Introduction

Polyethylene terephthalate (PET) is a significant source of micro-, nano-, and macroplastic pollution. While traditional plastic recycling is possible with PET, it can be extremely difficult to recycle all of the waste effectively and in an environmentally safe manner in its own right (1). Synthetic biology and metabolic engineering provide a solution by wrangling natural organisms to produce PET-degrading enzymes (natural proteins that catalyze chemical reactions) known as PETases. PETases can target PET bonds and break them down more efficiently than conventional industrial processes, and can be cleanly produced through fermentation. The challenge of synthetic biology is designing protein sequences that fold into efficient enzymes, a monstrously complex process. Deep learning, however, can handle the complexity of peptide chemistry and its emergent properties to an unprecedented degree (2). PETalytics aims to use deep, semi-supervised learning on a massive, sparsely labelled dataset to predict PETase properties, primarily enzyme activity, without the need for unnecessary wetlab characterization on inactive enzymes. PETalytics will work on top of existing state-of-the-art models and brand-new datasets with previously unseen scale in the hopes of outperforming previous models and techniques.

Background & Related Work

Protein Language Models (pLMs) use the similarities between human language and amino acid sequences through their common feature of complex context in one part of the input, having the ability to fundamentally modify the “meaning” of some snippets of data, and others none at all (3). ESM-2 (4) is one such transformer-based, self-supervised pLM trained to predict random amino acid residues masked from its input sequence. As a result, it began to interpret the relationship between different amino acids over large distances, and eventually learned to “see” physical residue-residue contact regions between parts of the protein in its attention patterns through this “understanding” (4). The resulting data expressed by a pLM before unembedding has been shown to represent rich information about the input sequence on its own through UMAP clustering (5) and when fed into further networks trained for a specific task, such as in the case of ESMFold: to find protein structures from single sequences (4). Muir *et al.* (6) showed that while fine-tuning these large, general pLMs is a common choice when creating specialized models, probing from zero-shot pLM embeddings with simple, feed-forward ANNs (MLPs) was better at predicting enzyme activity from a limited set of labelled training data. However, Muir *et al.* could not perform semi-supervised training in their experiments due to the dataset used, limiting their models. Variational Auto-Encoders (VAEs) are similar to pLMs in the sense that they are self-supervised and create latent vectors that encode compressed information about their inputs, but are much smaller and function much closer to a conventional ANN than a transformer (7). VAEs have been shown by Sinai *et al.* (8) to work well in protein-based regimes and protein design. Norton-Baker *et al.* (9) used a combination of VAEs and other machine-learning models to find new PETases, but notably trained their models directly on one-hot protein encoding and used their models to suggest PETases for wetlab characterization, rather than predicting properties in silico.

Data Processing

PETalytics will be built on the PETadex database (10), an open database made on over a billion sequences found by Logan (11) in the Sequence Read Archive (SRA). The database is largely raw sequences of unverified PETase-like proteins, but currently contains 231 protein sequences tested in the same assay for rich and detailed activity information. I plan to isolate the sequences of the 231 tested sequences and any other confirmed PETases in the PETadex for a separate training set. I will also cluster the database to varying degrees, as different clustering extents wildly change the diversity of the dataset for a given sample and change model fitting behaviour (4). I can also isolate sequences in the PETadex that are similar to confirmed PETases from the literature for fine-tuning rounds (9).

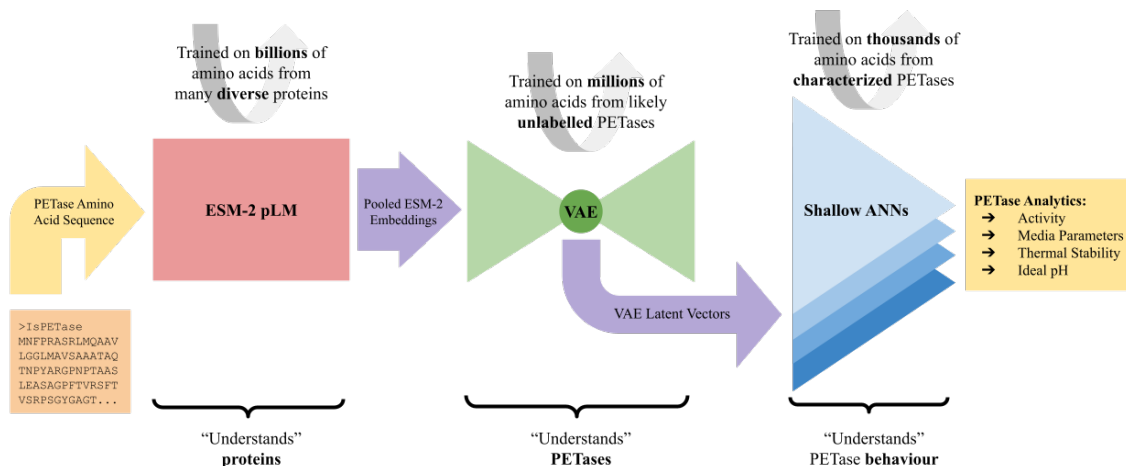


Figure 1: Overall model architecture summarizing the role of each stage of PETalytics and the data they are trained on.

Architecture

Visually described in Figure 1. I will generate pooled embeddings (either mean or min pooled according to (6)) of unlabelled PETases from the PETadex with the pretrained ESM-2 (4) and feed them into a VAE that will hopefully extract PETase-specific information from the embeddings into its latent space through self-supervision. After VAE training and validation, I will feed the latent vectors for PETases with known activity into shallow nonlinear ANNs (6) for supervised training on PETase analytics. The main hyperparameters I will be dealing with outside of the usual activation functions, learning rate, batch size, and epochs are the KL divergence weight in the VAE training, the dimension of the latent space of the VAE, its layers/layer sizes, and the layers/layer sizes of the prediction ANN.

Baseline Model

I will use a shallow, nonlinear ANN with one-hot encoded sequence data as my baseline model. I will also try zero-shot ESM-2 embeddings also fed into a shallow, nonlinear ANN similar to (6).

Ethical Considerations

The history of the PETadex dataset, going from the SRA to Logan to PETadex may have accumulated invisible biases from previous researchers on the inclusion and exclusion of certain datapoints. While the effects are largely unknowable, I will do my due diligence to spot and reduce biases I discover as well as try to avoid adding to them as I clean the data. The end-model itself also will not, by default, consider the specificity and potential toxicity of the enzyme to other organisms, however unlikely. Care must be taken to evaluate the safety of any PETases identified, and with more labelled data on PETase interactions with other organisms, PETalytics may help inform this.

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